

EuroSAT Land-Use Classification with CNNs and Vision Transformers

SYSC 5108

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Abstract

This project studies satellite land-use classification on the EuroSAT RGB dataset using three image classifiers: ResNet18, ResNet50, and ViT-Small. Each backbone was trained both from scratch and from ImageNet-pretrained weights under the same deterministic 70/15/15 train/validation/test split. The experiments focus on three questions: how CNNs compare with a vision transformer on a modest remote-sensing dataset, how much transfer learning helps, and how model size relates to accuracy and runtime. The saved results show a clear and consistent effect from pretraining. ResNet50 with ImageNet pretraining was the strongest configuration, reaching 98.67% test accuracy, 98.62% macro F1, and a test loss of 0.0458. The pretrained ViT-Small was close behind at 98.54% test accuracy, while the from-scratch ViT-Small was the weakest run at 90.40%. Overall, the experiments indicate that transfer learning is the dominant factor in this project, and that a pretrained CNN remains the most reliable option for this EuroSAT setting.

Table of Acronyms

CNN	Convolutional Neural Network
RGB	Red, Green, Blue image channels
GPU	Graphics Processing Unit
ViT	Vision Transformer
F1	Harmonic mean of precision and recall

Table of Math Notations

x	Input RGB satellite image patch.
y	Ground-truth land-use label.
$\hat{y}$	Predicted class-probability vector.
$f_{\theta}(x)$	Model with parameters θ .
$\mathcal{L}_{\text{CE}}$	Cross-entropy loss used for training.

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Chapter 1

Introduction

Satellite imagery is a practical source of information for land-use mapping, agricultural monitoring, environmental assessment, and urban planning. These applications benefit from automated classification systems because manual inspection does not scale to large image archives. Deep learning is a natural candidate for this problem because it can learn discriminative spatial features directly from image patches instead of depending on handcrafted descriptors.

This report focuses on the EuroSAT RGB dataset, a balanced benchmark of 27,000 Sentinel-2 image patches across ten land-use classes [1]. The project investigates three related questions. First, how do CNN baselines compare against a vision transformer on the same task? Second, how much does ImageNet pretraining improve performance? Third, is a larger model always worth its extra compute cost on this dataset?

The objective is not to claim a new state of the art, but to produce a clear end-to-end experimental study with reproducible training, validation, and test results. The final outcome is a compact comparison across six runs together with conclusions about convergence speed, class-level behavior, and the effect of transfer learning.

Chapter 2

Background

2.1 Remote-Sensing Image Classification

EuroSAT was introduced as a benchmark for land-use and land-cover classification using Sentinel-2 imagery [1]. The dataset provides ten balanced classes and reports strong CNN baselines, including an overall accuracy of 98.57% in the original paper. That published number gives a useful reference point for this course project, although the split protocol and implementation details differ, so only broad comparisons are appropriate.

2.2 Selected Deep Learning Techniques

Residual networks are widely used visual backbones because skip connections make deeper models easier to optimize [2]. In this project, ResNet18 serves as a lightweight CNN baseline and ResNet50 acts as the higher-capacity CNN.

Vision Transformers adapt the transformer architecture from sequence modeling [3] to image classification by converting an image into a sequence of patches [4]. Their strength is global token mixing, but their data efficiency is often weaker than strong CNNs when training from scratch. This makes ViT-Small an informative comparison point for a moderate-sized remote-sensing dataset such as EuroSAT.

All pretrained runs start from ImageNet weights [5]. Transfer learning is especially relevant here because EuroSAT is much smaller than large-scale natural-image corpora, so a pretrained backbone may converge faster and generalize better than a randomly initialized network.

2.3 Architectural Details of the Chosen Models

The three backbones differ not only in parameter count, but in how they process spatial structure. Both ResNet models begin with a 7×7 convolution with stride 2, followed by a 3×3 max-pooling layer. After this stem, they stack four residual stages and finish with global average pooling plus a linear classifier adapted to ten EuroSAT classes.

ResNet18 uses the standard $[2, 2, 2, 2]$ stage layout with **BasicBlock** residual units and a final $512 \rightarrow 10$ classification head. ResNet50 keeps the same four-stage pattern but expands it to $[3, 4, 6, 3]$ **Bottleneck** blocks, producing a $2048 \rightarrow 10$ output layer. This deeper bottleneck design increases representational capacity, but it also raises optimization difficulty when the model is trained from scratch on a relatively small dataset.

The ViT-Small model is instantiated as `vit_small_patch16_224`. In this project, the EuroSAT images remain 64×64 , so a patch size of 16 produces only $4 \times 4 = 16$ image patches. These tokens are projected into a 384-dimensional embedding space, prepended with one class token, and processed by 12 transformer encoder blocks with 6 attention heads and an MLP

ratio of 4.0. The final classifier is a $384 \rightarrow 10$ linear layer. The small number of patches makes the transformer computationally manageable here, but it also means that each token represents a fairly large spatial region of the original image.

Table 2.1: Concrete architecture details of the three backbones used in code.

Model	Main stem or tokenizer	Core depth	Task-specific head
ResNet18	7×7 conv + max-pool	2, 2, 2, 2 BasicBlocks	$512 \rightarrow 10$
ResNet50	7×7 conv + max-pool	3, 4, 6, 3 Bottlenecks	$2048 \rightarrow 10$
ViT-Small	16×16 patch embedding	12 transformer blocks, 6 heads	$384 \rightarrow 10$

2.4 Why These Models Were Chosen

The selected models cover two complementary design families and a meaningful range of parameter counts. ResNet18 is small and efficient, ResNet50 is deeper and more expressive, and ViT-Small has a parameter budget similar to ResNet50 while relying on a different inductive bias. This combination makes it possible to separate three effects: architecture family, model capacity, and pretraining.

Chapter 3

System Description

3.1 Dataset and Split

The experiments use the RGB version of EuroSAT with ten classes and 64×64 inputs [1]. A deterministic stratified split assigns 70% of the images to training, 15% to validation, and 15% to testing. Because EuroSAT is balanced, each class contributes 1,890 training images, 405 validation images, and 405 test images.

Table 3.1: EuroSAT RGB class distribution under the project split.

Class	Train	Validation	Test	Total
AnnualCrop	1890	405	405	2700
Forest	1890	405	405	2700
HerbaceousVegetation	1890	405	405	2700
Highway	1890	405	405	2700
Industrial	1890	405	405	2700
Pasture	1890	405	405	2700
PermanentCrop	1890	405	405	2700
Residential	1890	405	405	2700
River	1890	405	405	2700
SeaLake	1890	405	405	2700
Total	18900	4050	4050	27000

3.2 Preprocessing and Training

All models use the same image normalization statistics and the same data augmentation policy: random horizontal flips, random vertical flips, and random rotations up to 180° . The optimization setup is also fixed across runs: AdamW [6], a learning rate of 3×10^{-4} , weight decay of 10^{-4} , a batch size of 256, and 40 epochs.

The training objective is standard multi-class cross-entropy:

$$\mathcal{L}_{\text{CE}} = - \sum_{c=1}^C y_c \log \hat{y}_c.$$

Model selection is based on the highest validation accuracy, and the corresponding checkpoint is evaluated once on the held-out test set. For all six runs, the final classification layer was replaced so that each backbone predicts the ten EuroSAT classes. This keeps the feature extractor architecture intact while aligning the output space to the project task.

Table 3.2: Shared training configuration for all six runs.

Setting	Value
Dataset	EuroSAT RGB
Input size	64×64
Train/val/test split	70/15/15 stratified
Epochs	40
Batch size	256
Optimizer	AdamW
Scheduler	Cosine annealing
Loss function	Cross-entropy
Device	RTX 5080 GPU
Metrics	Accuracy, macro F1, loss, confusion matrix

3.3 System Pipeline



Figure 3.1: End-to-end pipeline used in the project.

Chapter 4

Numerical Results

4.1 Overall Comparison

Table 4.1 summarizes the six completed training runs. The strongest overall configuration was ResNet50 with ImageNet pretraining, which slightly exceeds the 98.57% EuroSAT benchmark reported in the dataset paper [1]. That said, the comparison should be interpreted cautiously because the exact split and implementation are not identical.

Table 4.1: Overall comparison across the six completed runs.

Run	Params (M)	Test Acc. (%)	Macro F1 (%)	Test Loss	Train Time (s)
ResNet18 Scratch	11.18	95.75	95.62	0.1362	34.52
ResNet18 Pretrained	11.18	98.42	98.36	0.0516	34.65
ResNet50 Scratch	23.53	93.16	92.91	0.2202	76.02
ResNet50 Pretrained	23.53	98.67	98.62	0.0458	75.29
ViT-Small Scratch	21.60	90.40	90.08	0.3038	58.17
ViT-Small Pretrained	21.60	98.54	98.48	0.0520	57.54

Two trends are immediate. First, every pretrained model outperformed its from-scratch counterpart by a large margin. Second, the from-scratch ViT-Small was the least data-efficient model in the study, while the pretrained ViT-Small nearly matched the best CNN.

4.2 Training Dynamics

Figure 4.1 shows that pretraining changes convergence behavior immediately. The pretrained ResNet50 reached at least 95% validation accuracy in the first epoch, the pretrained ViT-Small reached it by epoch 3, and the pretrained ResNet18 reached it by epoch 4. In contrast, none of the from-scratch models except ResNet18 ever crossed 95%, and the from-scratch ViT-Small never reached 90% validation accuracy within 40 epochs.

4.3 Accuracy, Capacity, and Transfer Learning

The parameter-count view in Figure 4.3 shows that larger models were not automatically better when trained from scratch: ResNet50 scratch was worse than ResNet18 scratch, and ViT-Small scratch was worst overall despite having almost as many parameters as ResNet50. Figure 4.4 makes the more important effect explicit. The largest gain from pretraining occurred for ViT-Small, which improved by 8.15 percentage points.

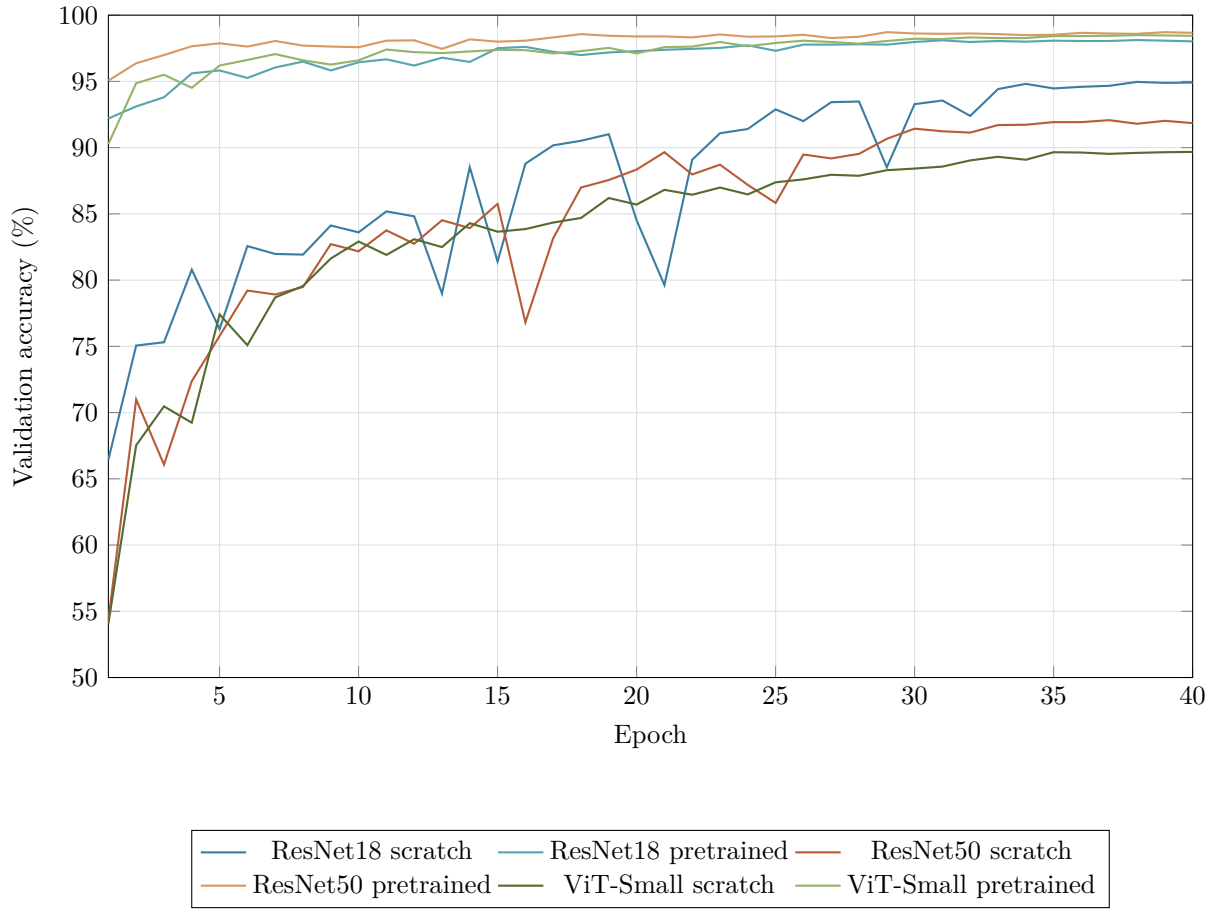


Figure 4.1: Validation accuracy over training epochs for all six runs.

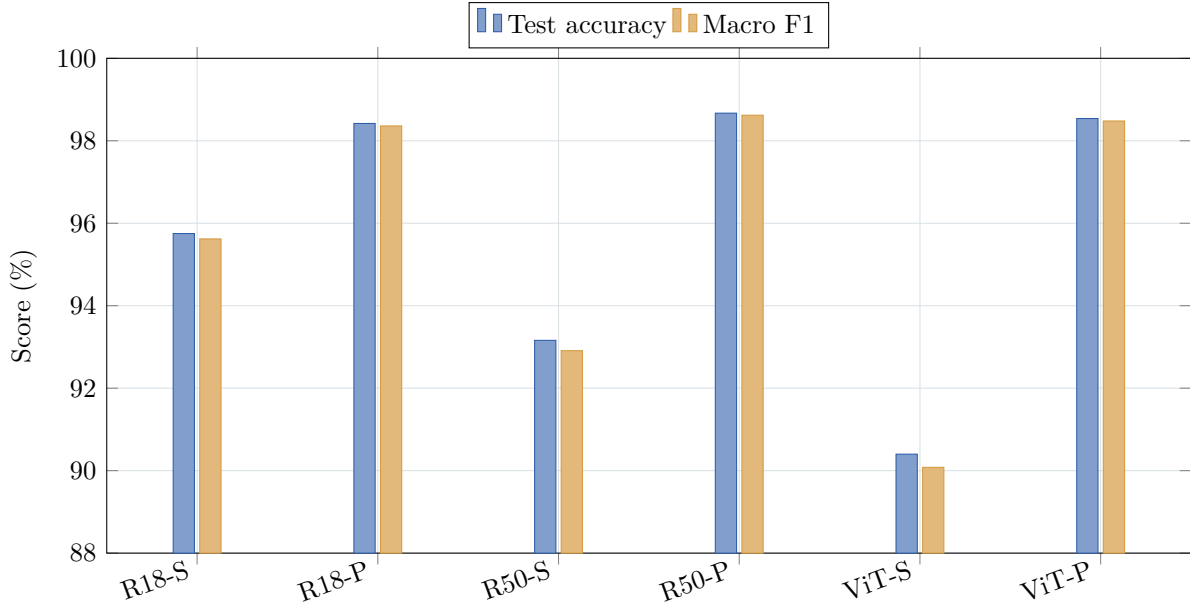


Figure 4.2: Test accuracy and macro F1 for each run. Tick labels use the abbreviations ResNet18 (R18), ResNet50 (R50), and ViT-Small (ViT), with S for scratch and P for pretrained.

4.4 Class-Level Behavior

Table 4.2: Hardest and easiest classes by per-class F1 score.

Run	Hardest class	F1 (%)	Easiest class	F1 (%)
ResNet18 Scratch	PermanentCrop	91.13	SeaLake	99.11
ResNet18 Pretrained	PermanentCrop	96.39	SeaLake	99.67
ResNet50 Scratch	PermanentCrop	86.83	SeaLake	99.00
ResNet50 Pretrained	AnnualCrop	97.14	SeaLake	99.78
ViT-Small Scratch	Highway	78.84	Forest	96.16
ViT-Small Pretrained	PermanentCrop	96.61	SeaLake	99.78

SeaLake was the easiest class in five of the six runs, while PermanentCrop was usually the hardest. This pattern is plausible because water bodies have strong visual structure, whereas PermanentCrop overlaps with AnnualCrop and HerbaceousVegetation more often.

The confusion matrix in Figure 4.5 is strongly diagonal, which matches the near-99% accuracy of the best runs. The remaining errors are concentrated in semantically similar classes. For example, the largest off-diagonal counts for the best model include PermanentCrop $\rightarrow$ AnnualCrop and Pasture $\rightarrow$ AnnualCrop, each with eight mistakes.

4.5 Attention-Map Batch Summary

The project also saved a small batch of ViT-Small attention visualizations. Since these figures already exist as report-ready raster outputs, it is more useful to show representative examples directly. Figure 4.6 includes two correct and two incorrect cases from the saved batch.

The attention montage makes two points clearer than the previous table-only summary. First, the correct examples show sharply localized focus on the dominant land-use pattern in the patch, which is consistent with the high-confidence predictions. Second, the incorrect cases

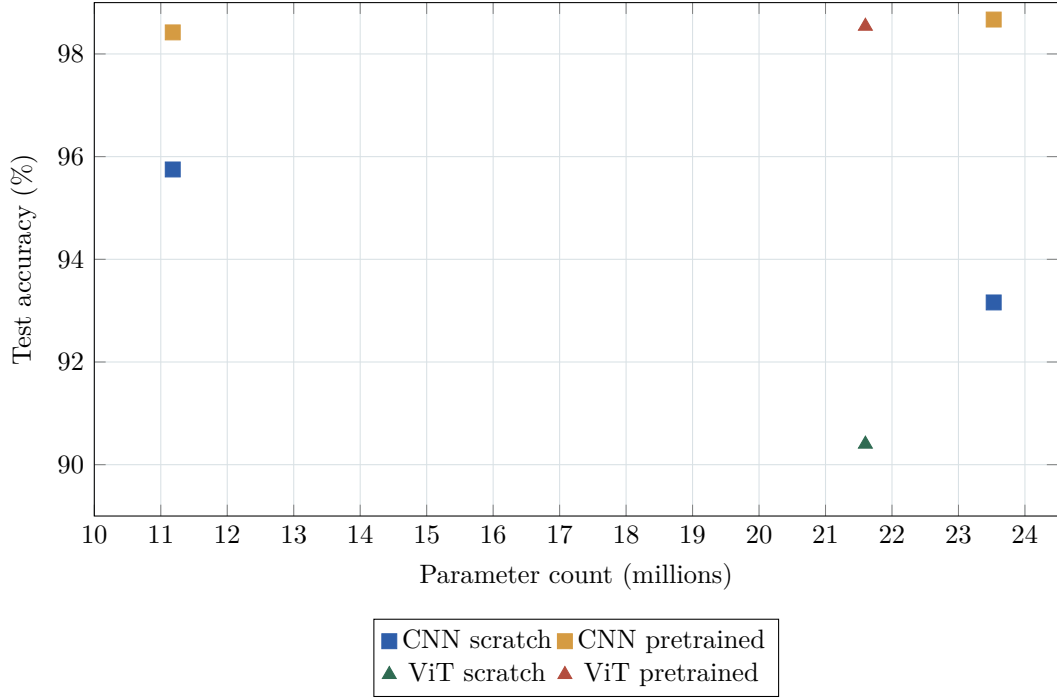


Figure 4.3: Model size versus test accuracy. The figure highlights that pretraining matters more than parameter count alone.

illustrate why some EuroSAT classes remain difficult: PermanentCrop and HerbaceousVegetation share fine-grained vegetation texture, while Residential and Industrial can overlap in roof pattern and road layout. The repeated PermanentCrop $\rightarrow$ HerbaceousVegetation confusion appears both here and in the full confusion matrices, so it is a dataset-level ambiguity rather than a one-off error.

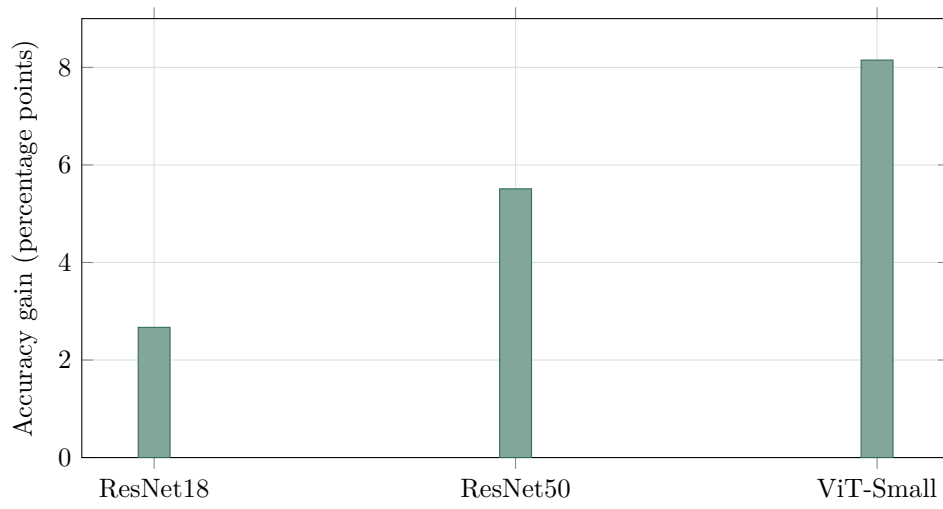


Figure 4.4: Test-accuracy improvement obtained from ImageNet pretraining.

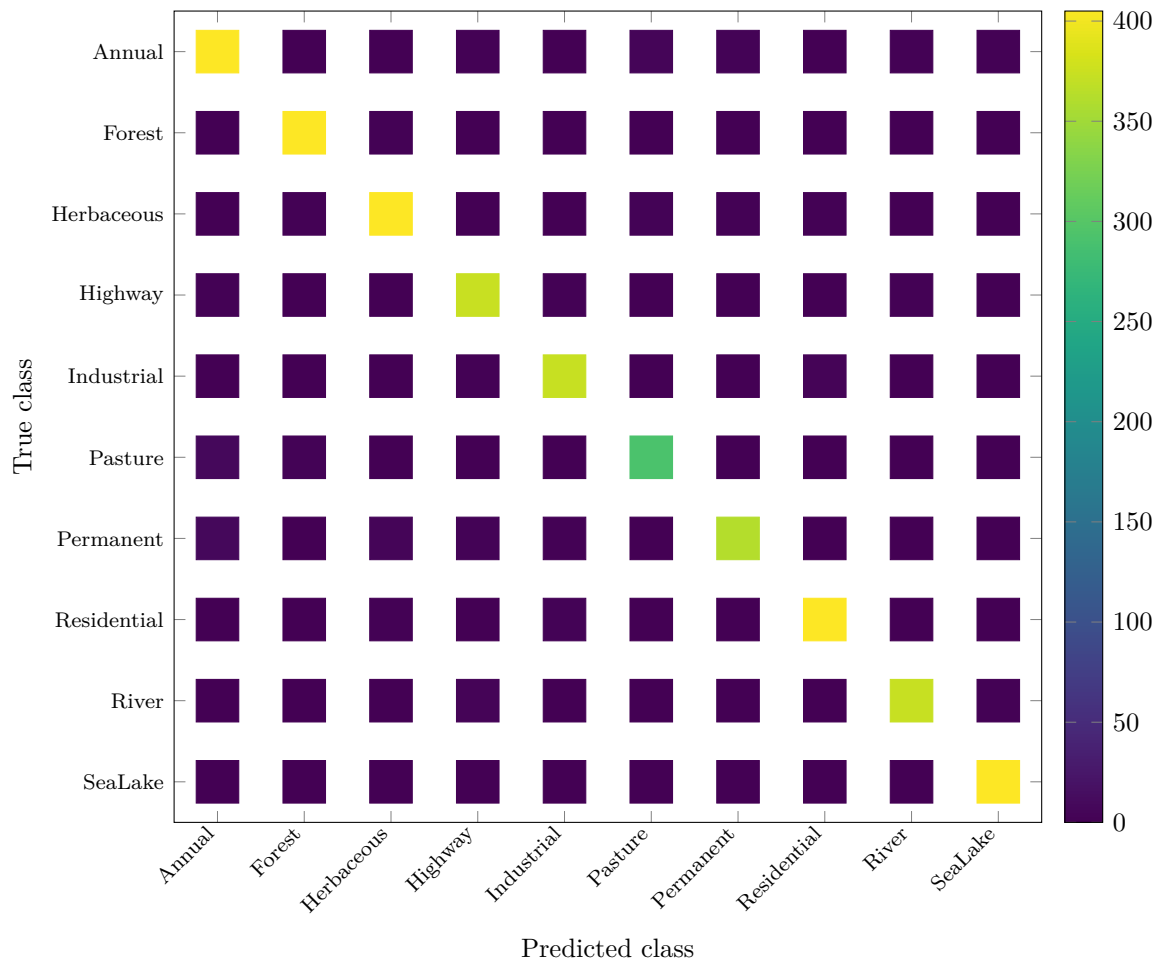


Figure 4.5: Confusion matrix for the best model, ResNet50 with ImageNet pretraining.

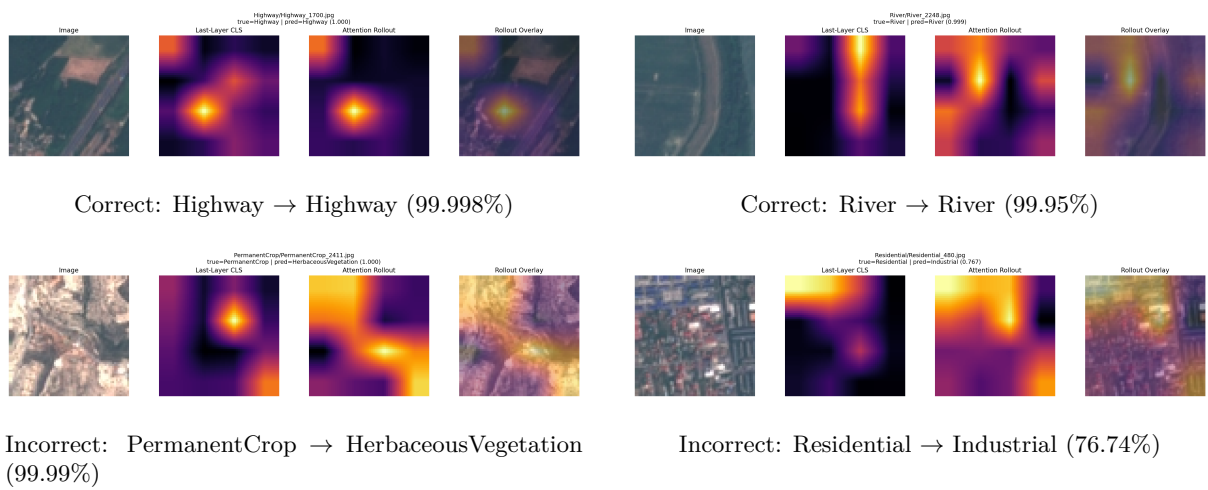


Figure 4.6: Saved ViT-Small attention visualizations from the test set. The bottom row shows that the model can still produce confident but wrong predictions when classes share similar texture or spatial layout.

Chapter 5

Conclusion

This project produced a complete comparison of six EuroSAT RGB classifiers under a fixed and reproducible training pipeline. The main conclusion is that transfer learning mattered more than architecture choice or raw parameter count. All three pretrained models reached at least 98.4% test accuracy, while their from-scratch counterparts lagged behind by 2.67 to 8.15 percentage points.

Among the tested configurations, ResNet50 with ImageNet pretraining was the strongest final model at 98.67% test accuracy and 98.62% macro F1. The pretrained ViT-Small was nearly as strong, which shows that transformer-based models can perform very well on EuroSAT once pretrained weights are available. However, the from-scratch ViT-Small result also shows that the transformer is the least data-efficient option in this specific setup.

The experimental evidence therefore supports a practical recommendation: for a small-to-medium remote-sensing dataset with limited project time, a pretrained CNN is the safest first choice, and a pretrained ViT becomes attractive when interpretability or architecture diversity is important. Future work could extend this report with multispectral EuroSAT experiments, robustness tests, or a larger transformer family.

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